$$I = \int_{\alpha}^{\beta} \left[\frac{\partial L}{\partial \boldsymbol{q}}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t))\boldsymbol{\eta}(t) + \frac{\partial L}{\partial \dot{\boldsymbol{q}}}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t))\dot{\boldsymbol{\eta}}(t) \right] \mathrm{d}t = 0.$$

We use integration by parts to remove the differentiation of $\boldsymbol{\eta}$, thus (in light of the boundary conditions on $\boldsymbol{\eta}(t)$),

$$I = \int_{\alpha}^{\beta} \left[\frac{\partial L}{\partial \boldsymbol{q}}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t)) - \frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{\boldsymbol{q}}}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t)) \right] \boldsymbol{\eta}(t) \mathrm{d}t = 0.$$

Since the variation $\boldsymbol{\eta}$ is meant to be arbitrary, it requires

$$\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{\boldsymbol{q}}}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t)) = \frac{\partial L}{\partial \boldsymbol{q}}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t)),$$

which is precisely the Lagrangian formulation of the equations of motion.

The method of derivation described here may be formulated in direct analogy to the traditional method of minimizing a function of several variables based on finding critical points. Define the *variational derivative* of a functional $\mathcal{F}$, $\delta\mathcal{F}/\delta\boldsymbol{q}$ so that

$$\mathcal{F}(\boldsymbol{q} + \varepsilon\boldsymbol{\eta}) = \varepsilon \frac{\delta \mathcal{F}}{\delta \boldsymbol{q}} \boldsymbol{\eta} + O(\varepsilon^2),$$

for all suitable (say, C^∞) functions $\boldsymbol{\eta}$. Viewing the action $\mathcal{A}_L(\Gamma)$ as the functional $\mathcal{F}$, the stationarity condition (the Euler-Lagrange equations) may be written

$$\frac{\delta \mathcal{F}}{\delta \boldsymbol{q}} = 0.$$

Thus the calculus of variations becomes a generalization of the traditional method of minimizing smooth functions by finding critical points.[4]

[4]This discussion is a great simplification. Any curve which satisfies this equation will represent a "stationary point" (actually, "stationary curve" would be more accurate) of the classical action. Such curves could include smooth local action minimizers, local action maximizers, or "saddle points" of the actional functional in a generalized sense. Deciding whether a given stationary curve is an actual minimizer of the action would require analysis of the second variation (the coefficient of ε^2 in the expansion above), which introduces additional complexity. For a more comprehensive treatment, see e.g. [210].